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Module Objectives
Prime Factorization
RSA Encryption and Decryption
RSA Key Setup
Video: RSA Key Setup
2 min
Video: RSA Example
4 min
Quiz: RSA Operations
Submitted
Security of RSA

RSA Operations

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Review Learning Objectives

1. What is the Euler Totient Function of 12, phi(12)? **3 / 3 points**

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Correct
See RSA Key Setup lecture.

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Due Apr 21, 11:59 PM IST Attempts 3 every 8 hours

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2. Using the same notations as in the lecture, e.g., p and q are the two primes and e and d are the public key and the private key, respectively, p=5, q=11, e=3. What is the value of phi(n) where phi(n) is the Euler Totient function of n and n is the modulus for RSA cipher? **3 / 3 points**

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40

Correct
See RSA Example lecture.

3. (This question builds on the previous question.) Using the same notations as in the lecture, e.g., p and q are the two primes and e and d are the public key and the private key, respectively, p=5, q=11, e=3. Using the phi(n) that you found in the previous question and the Extended Euclidean algorithm yields d=27. What is the ciphertext C when encrypting M=9? **3 / 3 points**

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Correct
See RSA Example lecture. You can also double-check by computing the decryption and comparing the result to the original message.

4. Using the same notations as in the lecture, e.g., p and q are the two primes and e and d are the public key and the private key, respectively, p=7, q=11, e=17. What is the value of phi(n) where phi(n) is the Euler Totient function of n and n is the modulus for RSA cipher? **3 / 3 points**

60

Correct
See RSA Example lecture.

5. (This question builds on the previous question.) Using the same notations as in the lecture, e.g., p and q are the two primes and e and d are the public key and the private key, respectively, p=7, q=11, e=17. Using the phi(n) that you found in the previous question and the Extended Euclidean algorithm yields d=53. What is the ciphertext C when encrypting M=8? **3 / 3 points**

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Correct
See RSA Example lecture. You can also double-check by computing the decryption and comparing the result to the original message.